Milestone 2: Dataset Preparation

Socratic AI Tutor

Group 2 | Data Science and AI Lab, IIT Madras BS

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What we prepared

Dataset	Used for	Size after preparation
ASSISTments 2009-2010	Learner model (DKT)	256,060 answers, 3,408 students, 82 skills
MathDial + SocraTeach	Socratic model fine-tuning	38,012 conversations
NCERT Maths + Science	Course pack (study text)	3,164 text chunks, 10 books

Scripts for every step are in docs/milestone-2/preprocessing/.

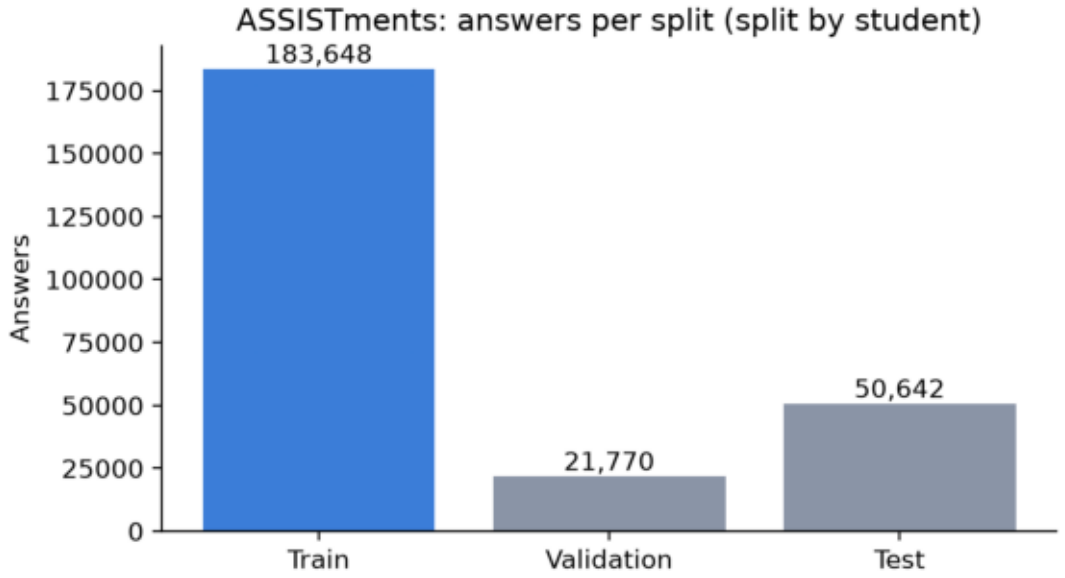
ASSISTments: learner model data

- 303,260 answers; kept 4 columns; removed 1 missing row, 46,668 duplicates, and students with under 3 answers

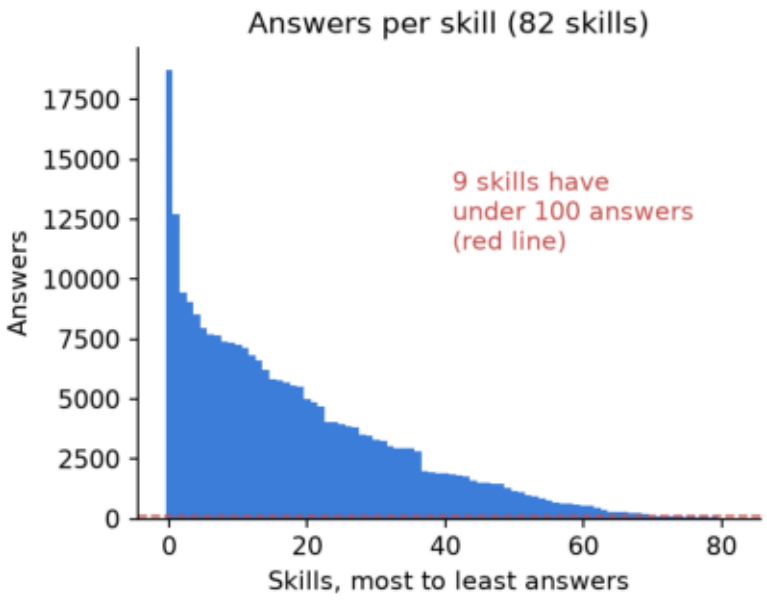
- 82 skills; each answer is one number: skill + 82 x correct

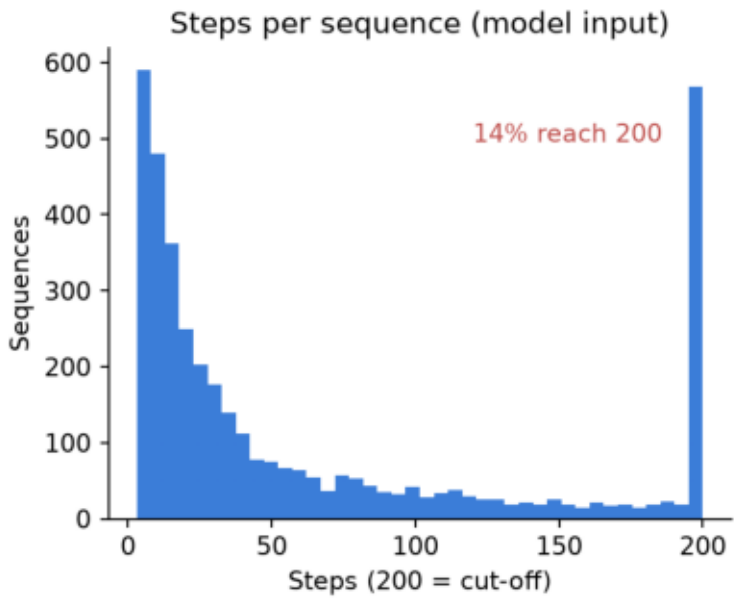
- Histories cut into chunks of 200, padded, with a mask for the padding

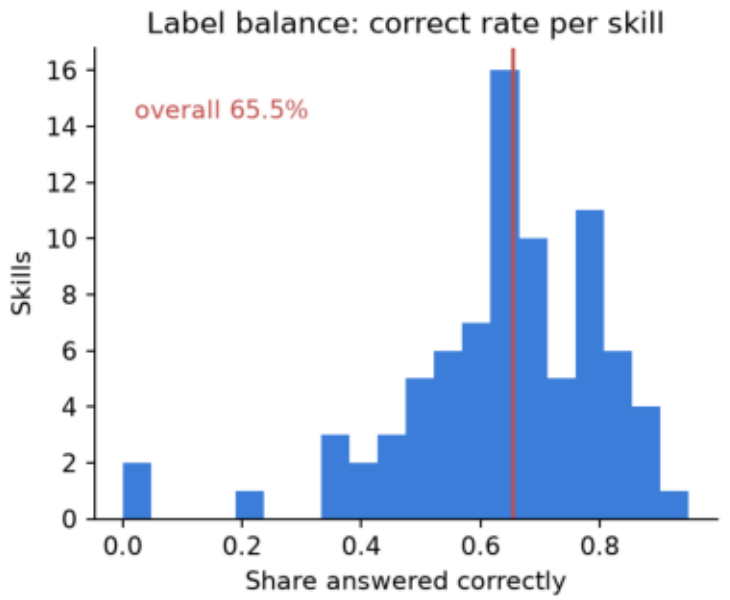
- Split by student: 80% train, 20% test; validation is 10% of the train students



ASSISTments EDA: skills, sequences and labels







- Skills are very uneven: the most practised skill has 18,728 answers while 9 of 82 skills have under 100. Results must be reported per skill, not as one number.

- Most sequences are short (median 30 steps) but 14% reach the 200-step cut-off, so long histories are cut into chunks.

- Labels: 65.5% of answers are correct, so a model that always says 'correct' scores 65.5%; the baseline to beat is a per-skill accuracy.

Chat corpus: Socratic model data

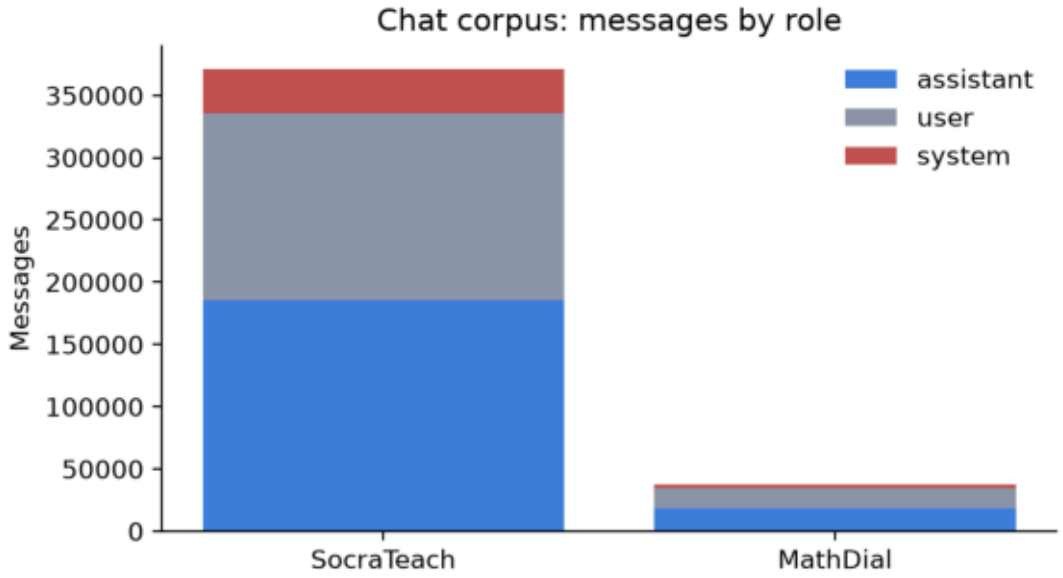
- 35,151 SocraTeach + 2,861 MathDial conversations

- Chat records: system, assistant and user messages

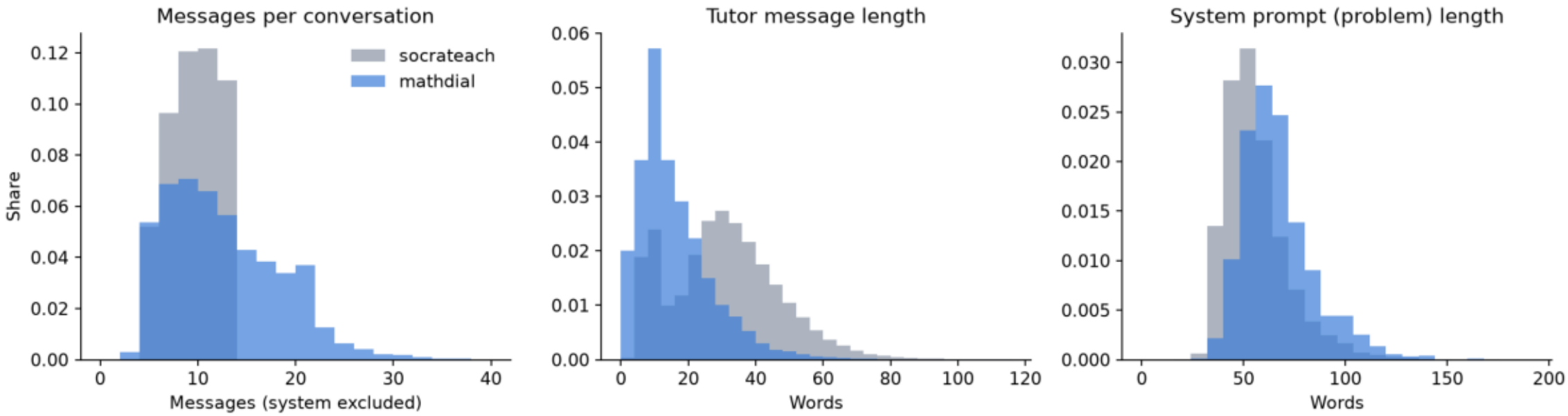
- MathDial rebuilt from the original release (no lost symbols)

- Validation and test mix both sources, split by problem

- Labels: tutor act and answer-leak flag



Chat corpus EDA: inputs

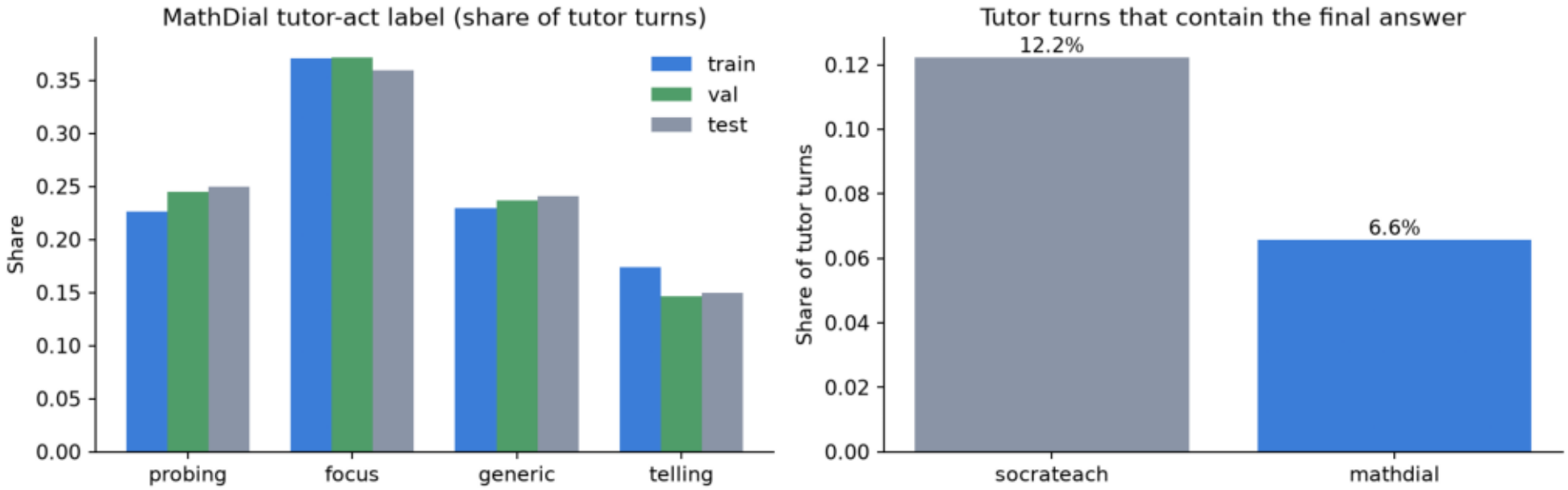


- SocraTeach is 92% of the conversations; its dialogues are shorter (median 9 messages against 11 for MathDial).

- MathDial tutor messages and problem prompts are longer: MathDial starts from a student's wrong solution, which is closer to our tutor's task.

- A model trained on the mix may lean toward SocraTeach's style, so we may weight MathDial and we report results per source.

Chat corpus EDA: labels

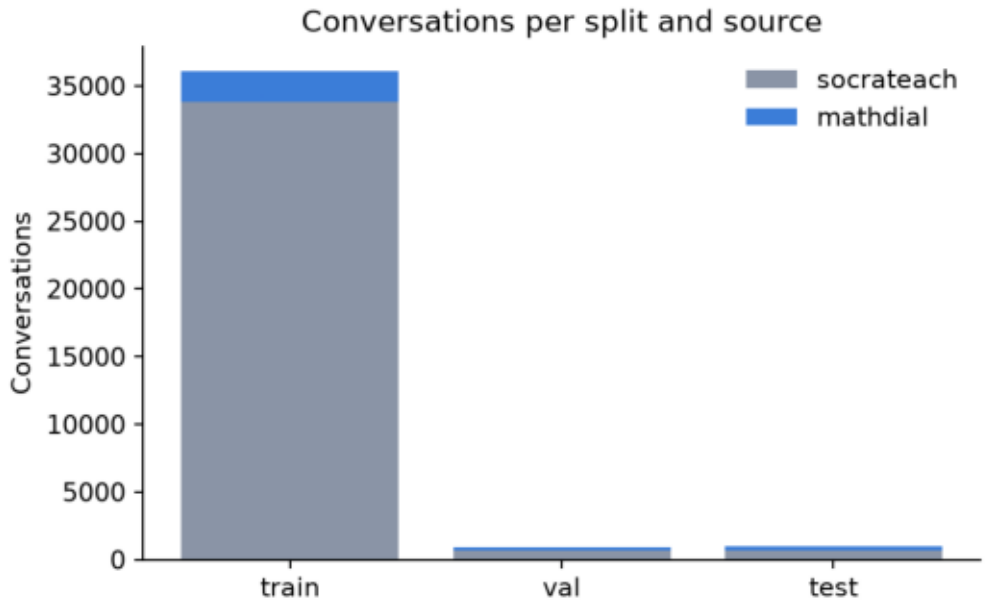


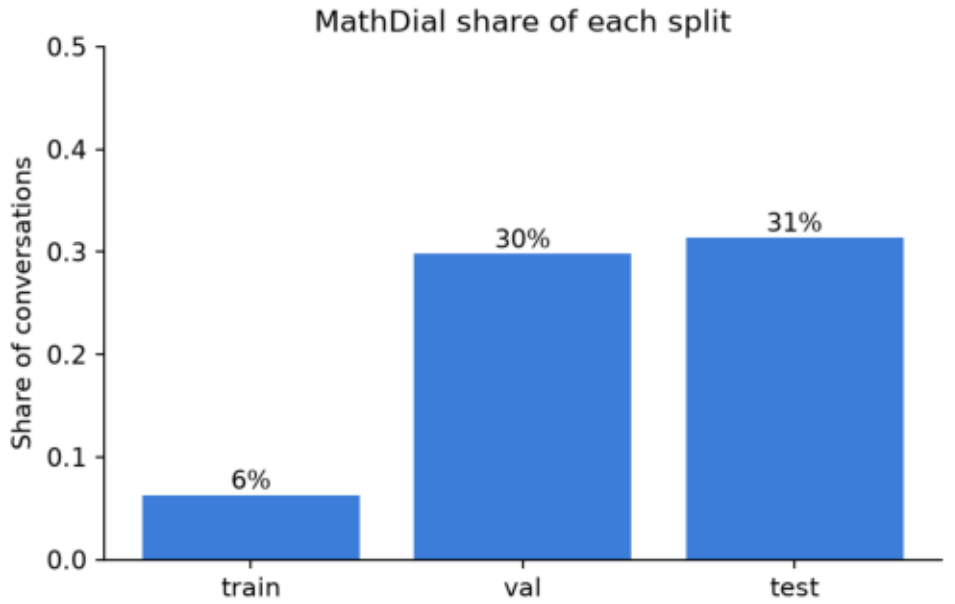
- MathDial tutor turns carry an act label: probing, focus, generic or telling. 17% are 'telling', which gives the method away; the model must not imitate them.

- Answer-leak flag: 12.2% of SocraTeach and 6.6% of MathDial tutor turns contain the problem's final answer (a number-match check, so short answers can match by chance).

- Both labels let us filter or down-weight training targets before fine-tuning.

Hybrid validation and test





- The same problem appears in both sources (1,006 of 1,035 MathDial training problems are also in SocraTeach) and in MathDial's own train and test sets (318 of 394).

- So conversations are grouped by problem across both sources, and a problem lives in exactly one split (checked).

- Validation and test each hold both sources, about 30% and 31% MathDial, so scores can be reported per source and combined.

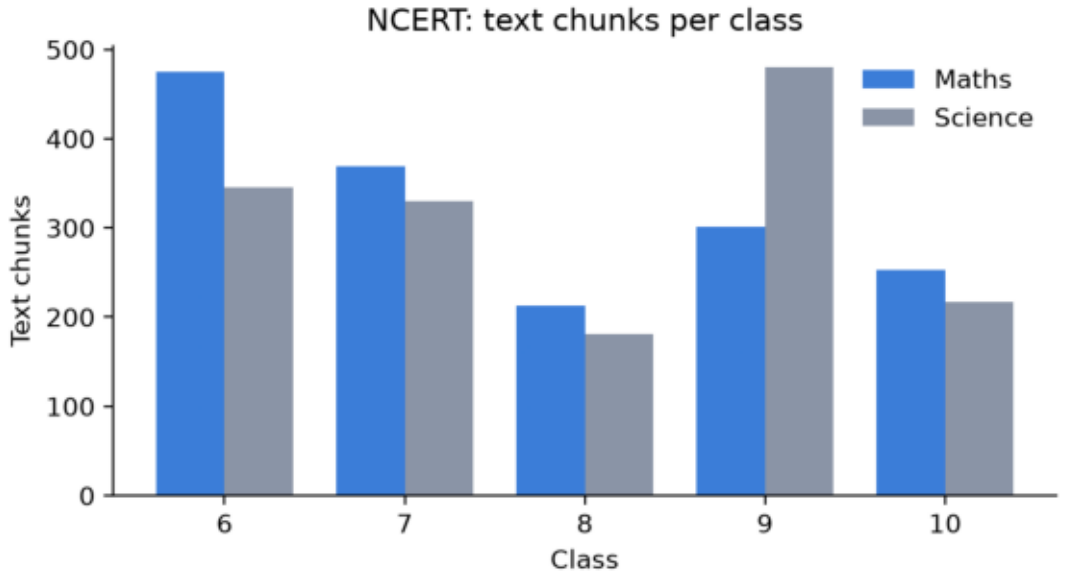
NCERT: study text

- Maths and Science, classes 6 to 10 (classes act as levels)

- 2,401 pages cleaned and cut into chunks of up to 250 words

- Each chunk keeps subject, class, chapter and page, so the tutor can point to the exact page

- Text stays local: the books are public but not ours to redistribute



Course packs: concepts and questions

- Hand-written chains of concepts that build across levels

- Example: percentages -> fractions -> multiplication and division

- Each concept links to its NCERT chapter, so the tutor can point to the page

- Practice questions have a hint and difficulty; hints never contain the answer

Pack	Concepts	Prerequisite links	Questions
NCERT Maths	10	13	52
NCERT Science	6	6	18

Our own test scenarios

- Backtracking: 82 cases (correct, careless, hint, gap one or two levels down)

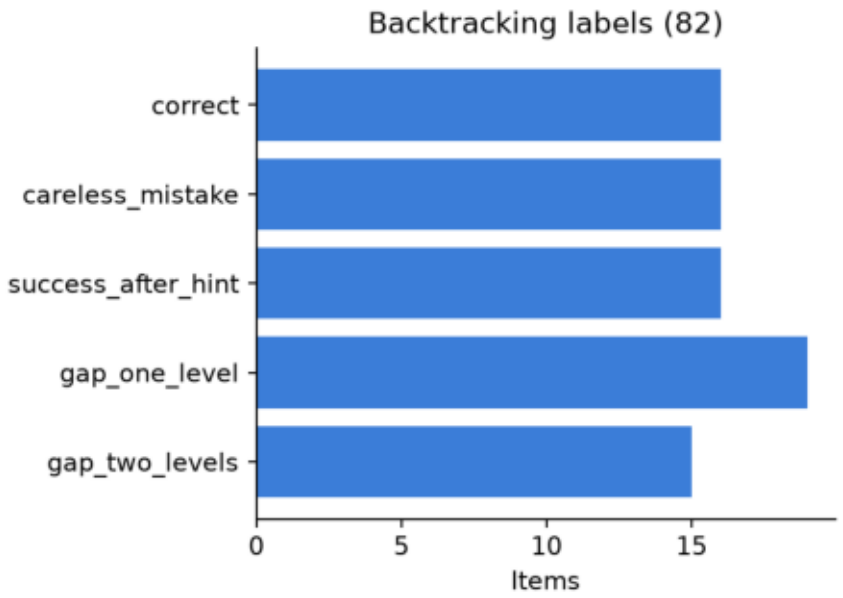
- Misconceptions: 32 wrong answers with the error behind each

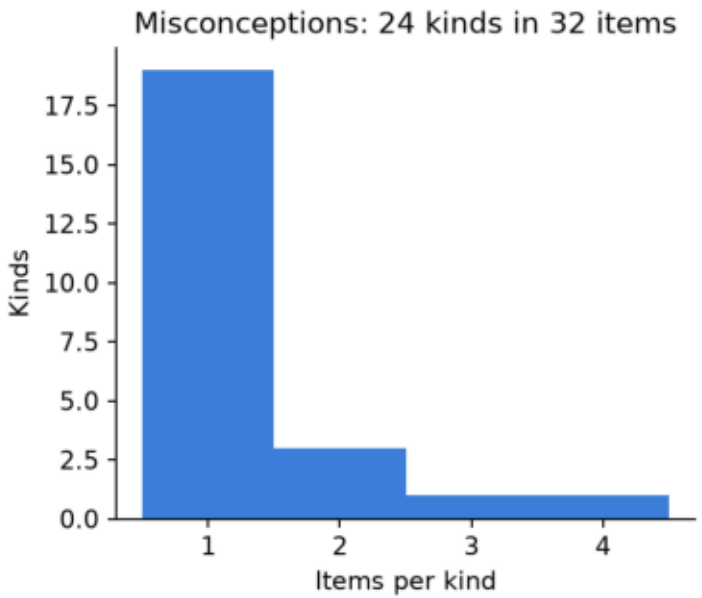
- Guardrails: 47 messages (on-topic, off-topic, answer request)

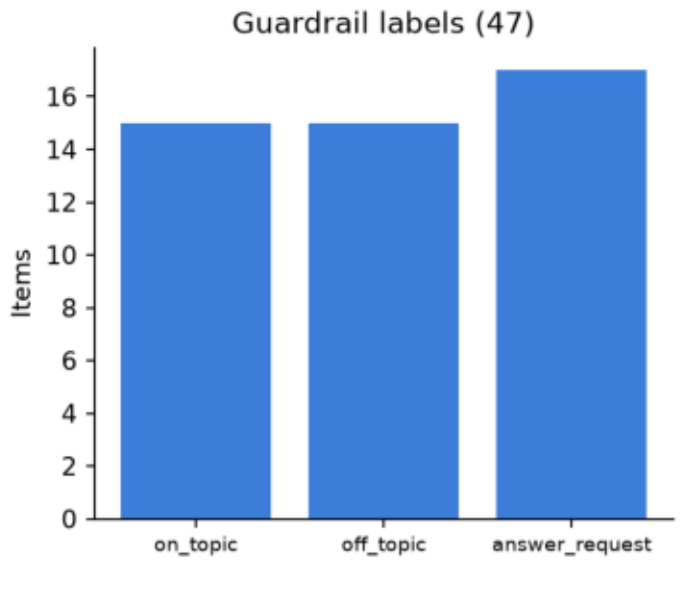
- Each backtracking case carries the learner's mastery, so cases with the same inputs always have the same label

- Never used for training or tuning; a second member reviews every label

Scenario labels and the dynamic misconception detector







- Labels are balanced across backtracking types (16 to 19 each) and guardrail classes (15, 15, 17); the 32 misconceptions cover 24 error kinds.

- The detector rebuilds wrong answers from each question's own numbers (13 error patterns plus a search over expressions), and remembers repeated errors per learner.

- Score: 31 of 32 scenarios named exactly, 31 of 31 on new numbers. Both sets are written by us, so real-learner testing is still to do.

Sample records

ASSISTments	`train,80959,chunk 0,step 1,skill 9,correct 1,x=91` (x = skill + 82 x correct)
Chat corpus	`{"source": "mathdial", "messages": [{"role": "system|assistant|user", "content": "..."}]}`
NCERT chunk	`{"subject": "maths", "class": 10, "chapter": 6, "page": 17, "text": "..."}`
Scenario	`{"type": "careless_mistake", "mastery": {"concept": 0.9}, "expected_first_action": "probe"}`
Misconception	`{"question": "What is 20% of 50?", "wrong_answer": "2.5", "misconception": "divides_by_the_percent"}`
Guardrail	`{"message": "What is the weather like today?", "label": "off_topic"}`

More samples and plots: Milestone2_EDA.ipynb

Processed dataset: folders and splits

```
hosted_dataset/
  assistments_dkt/    dkt_data.npz, dkt_processed_steps.csv, skill_map.csv
  socratic_chat/      train.jsonl, val.jsonl, test.jsonl
  splits/             dkt_validation.json
  ncert_metadata/     chapters.csv (text not shared)
  course_pack_specs/  specs.py
  test_scenarios/     backtracking, misconceptions, guardrails

ASSISTments: train 2,453 / validation 273 / test 682 students (by student)
Chat corpus: train 36,104 / validation 943 / test 965 (by problem, MathDial + SocraTeach in each)
```

Hosted dataset (Google Drive): https://drive.google.com/drive/folders/15vIRFghBsqfHjvHrIztgvVSUv2vP4tws

Checks and limits

- ASSISTments checks: row counts add up, encoding holds on every step, no student in two splits

- Our checks pass: NCERT chunks, pack graphs, no hint contains its answer, consistent scenario labels

- ASSISTments is US data with no session information

- SocraTeach is non-commercial (CC BY-NC); chat data is maths only

- NCERT text: some digits lost in older books; study passages need a manual pass

Status against the Milestone 1 plan

Plan	Status	Note
NCERT as primary course	Done	text, chunks, 2 course packs
IITM BS pack	Not done	pipeline is course-agnostic; add later
Guardrail methods compared	Not done	eval set built (47); methods in M3
Controller / misconception baselines	Not started	moves to M3
Class 3-5 content (level 1)	Not done	only classes 6-10 processed

Next steps

- Second member reviews concepts, prerequisites and scenarios

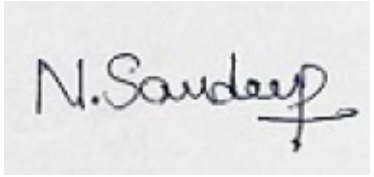
- Train DKT on ASSISTments; published baseline AUC 0.754 (pyKT benchmark, Liu et al. 2022)

- Add an answer-leak flag, then fine-tune the Socratic model on the chat corpus

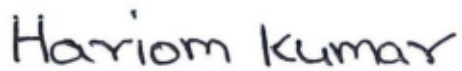
- Add the TA's test samples to our scenarios

Team sign-off

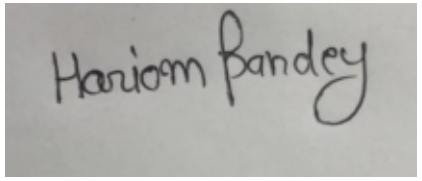
Sandeep Nalla (21f1006125)



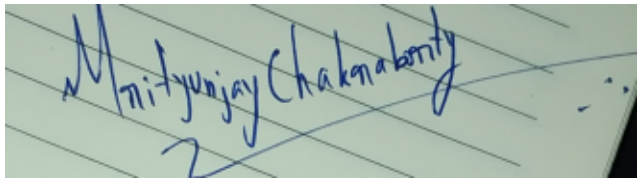
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Signed 29 October 2026